JalSakshi

Delhi pays for rainwater harvesting it cannot check. We measure it.

Team: ‹TEAM NAME› · **Team ID:** ‹ID› · **Theme 1:** IoT-based underground water management

‹Member 1 — role› · ‹Member 2 — role› · ‹Member 3 — role› · ‹Member 4 — role› · ‹email / phone›

A rule Delhi funds, but can't verify

What	Status	Source
RWH mandatory for plots ≥ 100 m²	in force **since 2001**	Indian Express (2026)
10 % water-bill rebate if functional (15 % with recycling)	in force	Hindustan Times, 21 Apr 2026
₹204.46 crore rebates paid, 2017–2024	reported	HT, citing DJB data
Functionality certificate	issued **after inspection**	HT, 21 Apr 2026
Annual owner self-certification	**being planned**	HT, 21 Apr 2026
1.5× bill for non-compliance	**"may attract"**	HT, 21 Apr 2026

> "Till now, there was a rule but no real checking." — Delhi water minister (HT, 21 Apr 2026)

"Many have become non-functional due to inadequate maintenance" — DJB official (Indian Express)

Why it matters now

- Delhi extracts **92.1 %** of its annually extractable groundwater. **10 of 34 tehsils** are over-exploited (337.44 km²).

  CGWB, *Dynamic Ground Water Resources of India 2025*, §7.32

- DJB Phase I: **restore 1,000 non-functional** RWH systems in government buildings and build 500 new ones (target 30 Jun 2026). 7,596 of 9,148 government buildings already have RWH.

  Indian Express (2026)

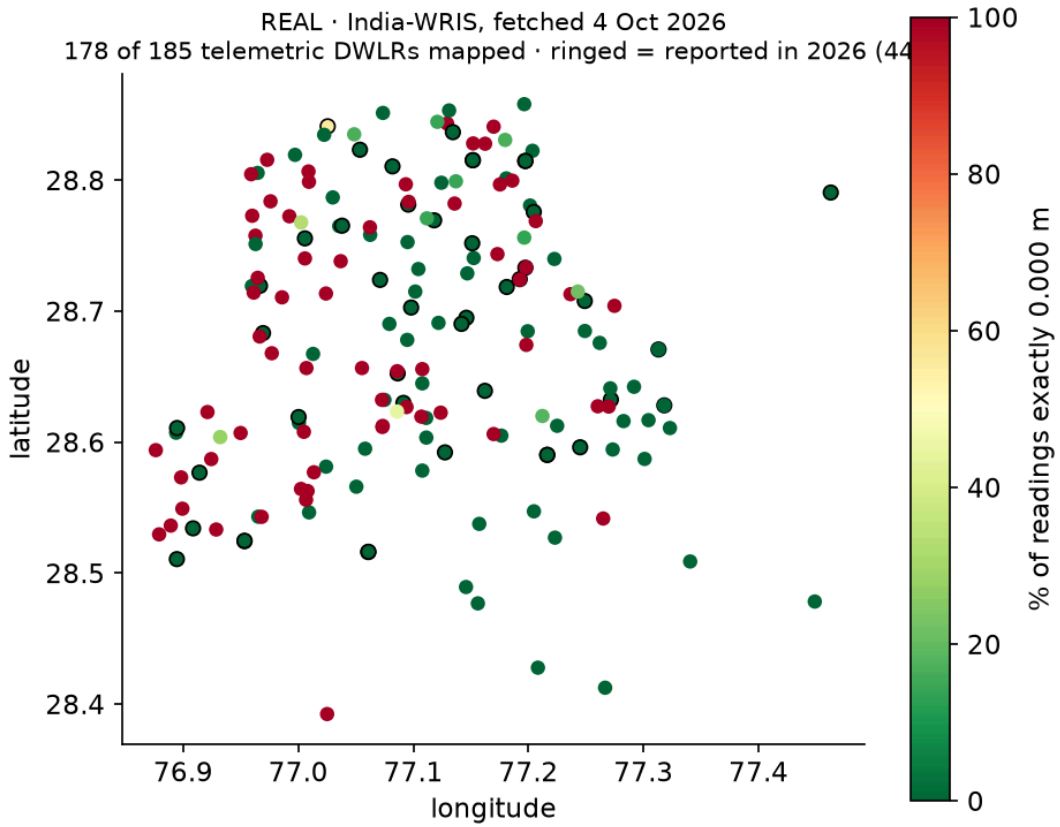
- DJB estimates the campaign's recharge potential at **≈ 733 million litres a year**. Today nobody can say how much of that reaches the ground.

  Indian Express (2026); an estimate, not a measurement

Even the city's own sensors need checking (our analysis, REAL data)

India-WRIS public API, fetched 4 Oct 2026 (483,253 Delhi readings):

- **185** telemetric piezometers (DWLR), reporting every 6 h
- only **46** reported anything in 2026
- **23 %** of readings are exactly 0.000 m; **17 %** of 6-h slots are missing
- where the data are clean, water levels rose a median **2.5 m** in the 2025 monsoon (79 stations)



Every number is sourced on its slide · REAL = public data · SIMULATED = labelled simulation

JalSakshi: one witness per recharge pit

In the pit: a 0–5 m stainless steel pressure probe (±0.5 % FS) in a slotted, socked stilling tube
In the lid: MCU + LoRaWAN radio, battery and a signed local log. Optional rain gauge.
Sampling: every 15 min when dry, every 1 min when rain or a rise wakes it

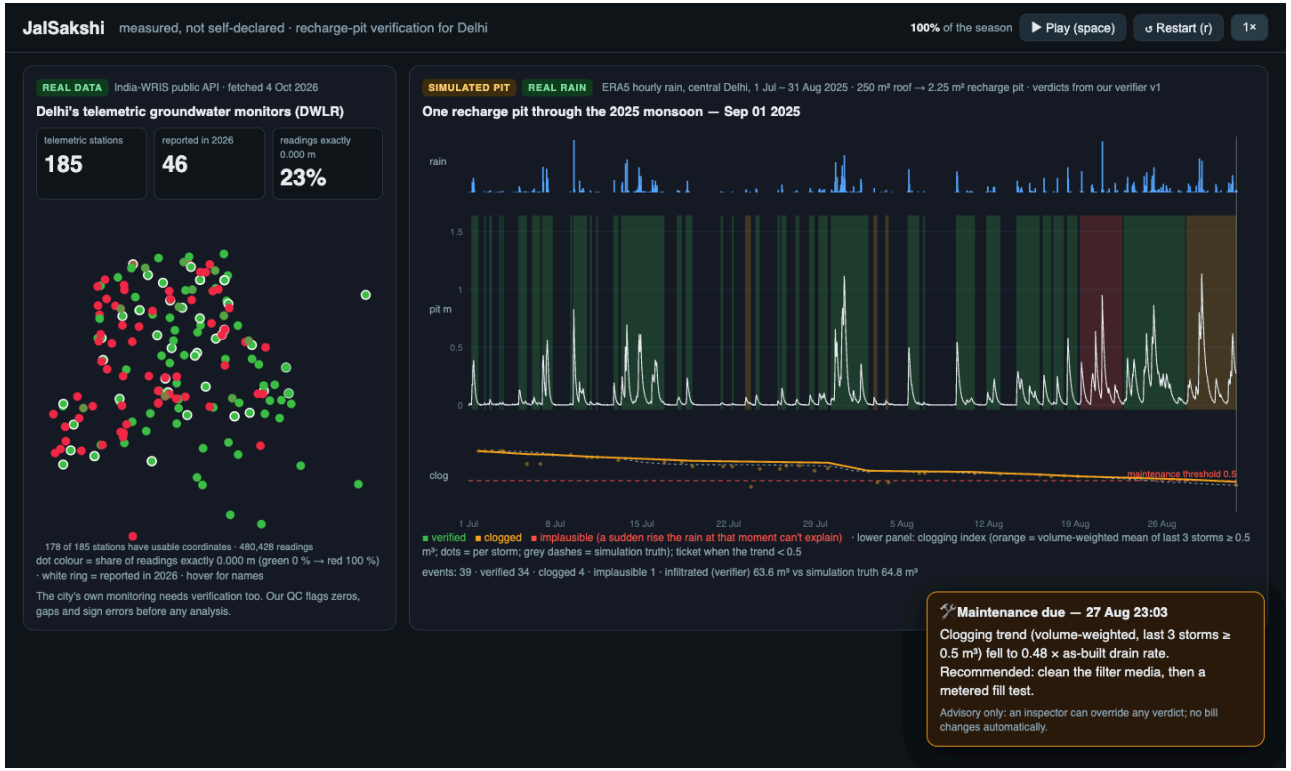
Two outputs per storm, nothing more:

1. **Drain rate** (the storm runs a falling-head test for us) → **clogging index** vs as-built
2. **Infiltrated volume ± band**: the rise is checked against rain × roof area, overflow is capped at the spill level, and one metered hose fill fixes pit area and drain rate

Verdict: verified · clogged · no inflow · implausible (a rise with no rain)
→ **ranks inspections; an inspector can override; no bill changes automatically**

60-second demo (offline laptop)

REAL Delhi DWLR network → **SIMULATED** pit on **REAL** 2025 rain → a verdict per storm → **"Implausible"** and **"Maintenance due"** tickets



How good is it? (simulation today; a blind rig test by 1 Dec)

Simulated pits, real Delhi rain; median error of infiltrated volume per storm.

Test (frozen; monsoon-2024 holdout)	Median error	1 storm in 10 is off by more than
Stress test: non-linear drains + imperfect rain gauge	**13.1 %**	58 %
+ realistic probe: ±25 mm offset, drift, quantisation	**14.5 %**	61 %
Same drain model as the verifier (best case)	3.0 %	16 %

- Fake pours (no rain to explain them) flagged: **10/10 simulated fakes** in every test. Honest storms wrongly flagged: ≤ 1.3 %.
- Clogged-pit verdict (end of season): precision 0.60–0.67, recall 0.90–1.0 (stress tests, holdout)

We **rejected** five candidate fixes under a rule set before running them: four helped one test but hurt another, and one didn't help enough.

Built for Delhi conditions

- **Underground link:** only the probe is below ground. The antenna sits flush with the lid. A local buffer means no lost events.
- **Accuracy:** ±25 mm per reading (datasheet) → ±0.11 m³ worst case on a 2.25 m² pit, about 3 % of a 4 m³ storm. A dry-floor auto-zero removes constant offset.
- **Silt:** probe on a stand above the silt line, inside a socked tube. A floor reading during rain raises a check.
- **Heat & water:** probe rated −20 to 70 °C, 316L stainless steel, IP68; IP67 enclosure.
- **Power:** probe switched on only to sample. **Battery life gets measured, not promised.** Settle time decides D-cell vs solar.
- **Calibration:** one metered hose fill at commissioning, an annual dip check, a re-fill after cleaning.

Probe: DFRobot KIT0139 datasheet (farnell.com/datasheets/3708809.pdf)

Cost, buyer, rollout

- **Node: ₹11,166 in verified retail parts** (probe ₹4,254.51 at element14; LoRa module, host board, Li-$SOCl_2$ cell, IP67 box; tube/cable/potting not yet priced)
  vs one CGWA-compliant DWLR (Acorn): **≈ ₹38,000**, logging twice a day
- **First buyer: DJB's Phase-I restorations of 1,000 government buildings.** DJB has **59 empanelled RWH service providers** who could install it, and is building a compliance-tracking portal our events could feed (Indian Express).
- **Then:** campuses and RWAs that want proof for the rebate.
- **Model:** node + annual verification service. Verdicts rank inspections. **₹204 cr is not our market.**

Impact and the ask

- **Audit DJB's own claim** (≈ 733 M L/yr) structure by structure, in measured litres
- **Clogging alerts** send cleaning crews to the pits that stopped working, before the monsoon
- **Data hygiene:** QC for the city's DWLR stream, so the aquifer response can be checked
- **Roadmap:**
  - Oct: drum pour rig + node
  - Nov: blind weighed-water trials + 1–3 real pits
  - 1 Dec: report + working prototype
- **Ask:** a pilot partner (DJB zone, campus or RWA), and permission to instrument 3 pits

Measured, not self-declared.

Backup slides

Backup 1: Why we don't claim to catch illegal borewells

- DJB identified **20,297** illegal borewells; **15,962** sealed; **4,000+** "couldn't be traced on the ground" (HT, 10 Jun 2025; Millennium Post). The NGT found seals that let borewells run "very conveniently" (The Bar Bulletin, 2026).
- Public piezometers report **every 6 h**, so pump cycles shorter than ~12 h alias. Drawdown is confounded by legal pumping, barometric pressure and rain.
- So we **never accuse a well**. A zone-level influence index is research only.

Backup 2: BOM (verified 4 Oct 2026, single-unit retail)

Part	Vendor	₹
DFRobot KIT0139 0–5 m 4–20 mA probe	element14 India	4,254.51
RAK3172 LoRaWAN module (IN865)	electronicscomp.com	1,672.80
ESP32-C6-DevKitM-1 (prototype host)	robu.in	1,099.00
Forte ER34615 D Li-$SOCl_2$ 3.6 V	robu.in	879.00
IP67 ABS box 150×100×70 mm	amazon.in	3,250.87
PG7 gland	robu.in	10.00
Subtotal		**11,166.18**
optional: tipping-bucket rain gauge (DFRobot)	amazon.in	3,729.00
shared: LPS8N 8-ch LoRaWAN gateway	amazon.in	33,559.00

URLs and access dates: deliverables/prototype-plan.md §3. Not priced: boost converter, antenna, tube, sock, potting, cable.

Backup 3: The evaluators (frozen, pre-registered keep rule)

- **v1:** linear-reservoir pits (same model as the verifier) → best case
- **v2:** drain outflow ∝ h^β, β ~ U(0.5, 1.5); rain gauge × lognormal(0.25)/day; 0–5 min lag
- **v3:** v2 plus a probe offset of ±25 mm, ±10 mm/season drift, readings floored at 0 m, 2.1 mm quantisation
- **Fleet:** 40 pits per split (healthy / clogging / blocked / faker); registry ±5–10 % wrong. Dev = monsoon 2025, holdout = monsoon 2024 (real ERA5 rain).
- **Keep rule:** improve the target by > 0.5 points and worsen no other evaluator by > 10 %; confirm on holdout.